Where's Fluffy
A Cloud-Based Pet Recovery Platform

Where's Fluffy is a cutting-edge platform designed to revolutionize pet recovery through advanced technology and community engagement. This presentation outlines the technical architecture, key features, and strategic vision behind our solution.

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Executive Summary

Problem: Fragmented pet recovery methods with low success rates

Solution: Cloud-based platform with GPS tracking, photo verification, and reward management

Architecture: AWS microservices with 99.9% uptime and <200ms response

Scale: Single city to multi-metropolitan expansion capability

Impact: Target 70% pet recovery rate through community coordination

Current Pet Recovery is Broken

Fragmented & Slow Process

Traditional recovery methods lack coordination, resulting in delayed reunions and frustrated owners.

Outdated Communication

Craigslist posts and paper fliers reach limited audiences with no real-time updates.

No Location Intelligence

Communities lack GPS-enabled platforms for immediate, location-location-based pet discovery.

Key Requirements: Foundational Pillars

Location-Based Discovery

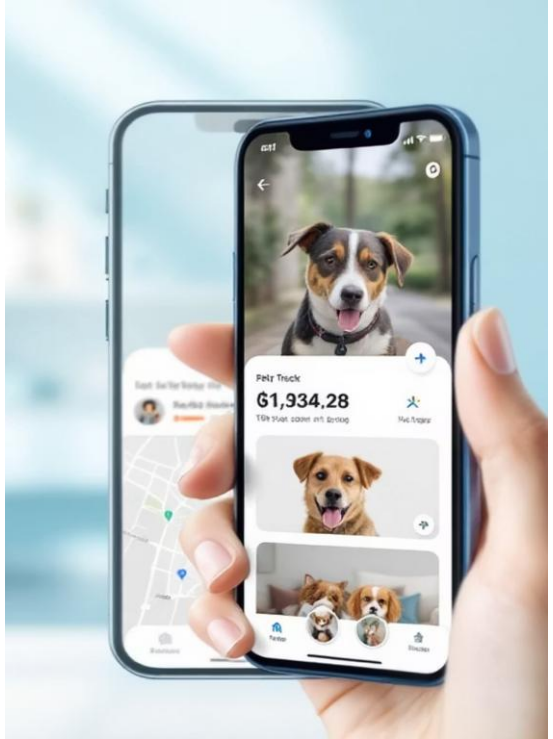
Pinpointing lost pets in real-time on an interactive map, leveraging precise geo-tagging.

Verified Photo Proof

Ensuring authenticity and reducing fraud through mandatory, timestamped photo submissions.

Escrow Reward System

Securely managing and distributing rewards upon verified pet recovery.



Mobile-First Design

Optimizing for urgent, on-the-go reporting and access via intuitive mobile interfaces.

Community Engagement

Fostering a collaborative network of pet lovers to aid in search efforts.

Stakeholders & Use Cases

Key Stakeholders

- **Pet Owners:** Users posting lost pet alerts and managing rewards.
- **Pet Finders:** Individuals submitting sightings and claiming rewards.
- **Community Members:** Volunteers sharing information and local alerts.
- **Admins:** Platform management, moderation, and support personnel.

Primary Use Cases

- **Post & View Lost Pets:** Owners create detailed profiles; others browse local listings.
- **Submit Sightings:** Finders report pet locations with photo evidence.
- **Confirm Found Reports:** Owners verify and finalize recovery processes.
- **Reward Distribution:** Secure release of funds upon successful pet reunion.

Solution Overview

1

Mobile + Web Platform

A unified interface across native mobile applications (iOS/Android) and a responsive web portal, ensuring accessibility for all users.

2

Real-time GPS Pet Tracking

Leveraging integrated mapping services and user-submitted data for live updates on pet locations and proximity alerts.

3

Mandatory Photo Proof

Strict requirement for photo submissions, enhanced with EXIF data analysis to verify location and timestamp integrity.

4

Reward Escrow with Stripe

Secure integration with Stripe Connect for holding and distributing rewards, ensuring trust and transparency in transactions.

5

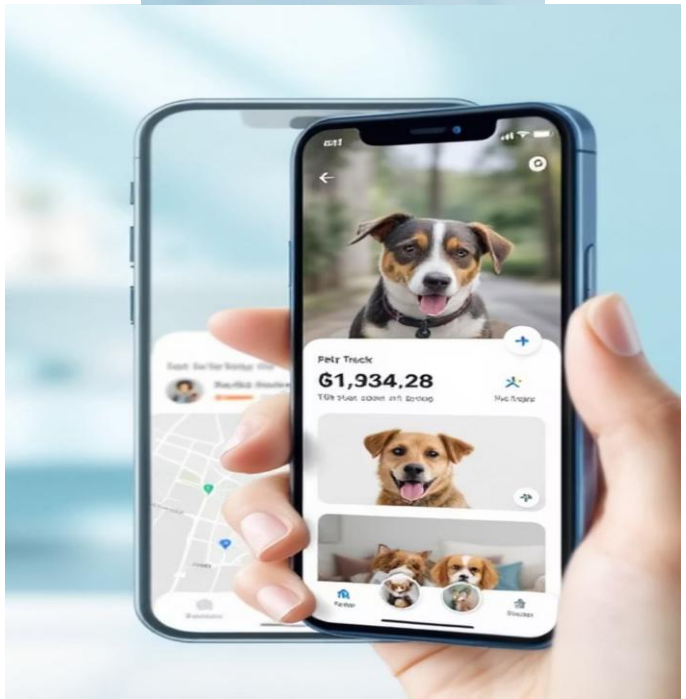
Community-Sourced Updates

Enabling users to contribute sightings, share alerts, and collaborate on pet recovery efforts in their local areas.

6

Notification & Comment Systems

Real-time push notifications and in-app messaging to facilitate rapid communication between pet owners and finders.

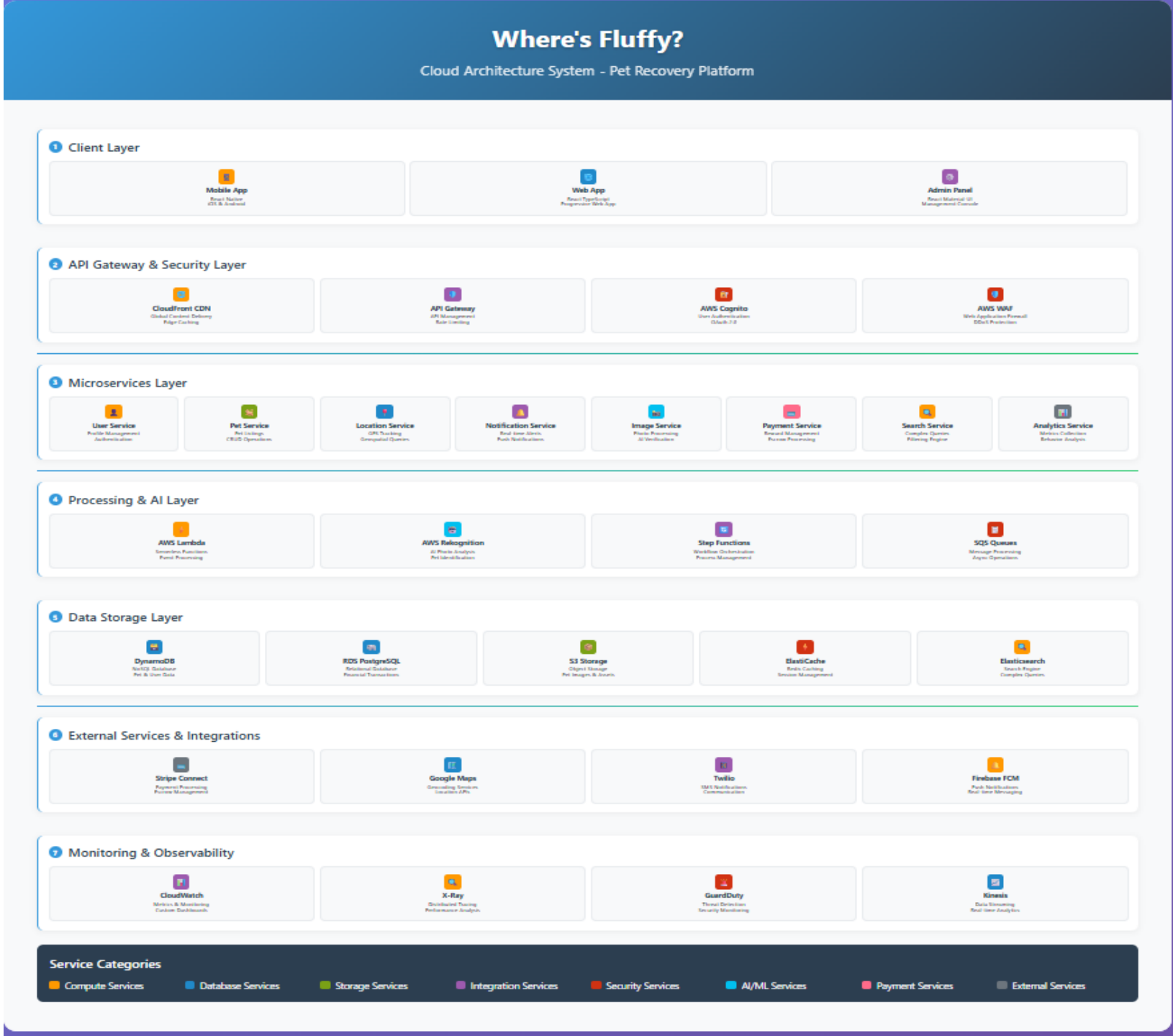


System Architecture: Microservices on AWS

"Where's Fluffy?" leverages a scalable microservices architecture hosted on Amazon Web Services (AWS) to ensure high availability and performance.

- **API Gateway:** Manages all incoming requests, routing them to appropriate microservices.
- **ECS & Lambda:** Containerized services (ECS) and serverless functions (Lambda) for core logic.
- **S3:** Secure and scalable storage for all user-uploaded pet photos and other static assets.
- **DynamoDB:** High-performance NoSQL database for flexible data storage (user profiles, pet details, sightings).

Link: [Architectural Diagram]



Data & Tech Stack

Data Persistence

- **DynamoDB:** Primary database for user profiles, pet listings, and sighting data. Optimized for high read/write performance.
- **PostgreSQL:** Relational database for financial transactions, audit logs, and complex reporting (e.g., reward payments).
- **Elasticsearch:** Powers advanced search functionalities for pets and sightings based on multiple criteria (breed, location, status).
- **Redis:** Utilized for caching frequently accessed data and managing user sessions to enhance responsiveness.

Technology Stack

- **React:** Frontend framework for dynamic and interactive web interfaces.
- **Node.js:** Backend runtime for building scalable microservices and APIs.
- **Python:** Used for data processing, analytics, and potential machine learning integrations (e.g., image recognition).
- **AWS Ecosystem:** Leverages various AWS services for infrastructure, compute, and storage.
- **Stripe:** Integrated for secure and compliant payment processing, specifically for the reward escrow system.
- **Google Maps API:** Provides mapping, geolocation, and routing services for accurate pet discovery.

Technology Stack

Frontend
React Native, React TypeScript, Redux Toolkit

Backend
Node.js, Express, Python, Socket.io

Databases
DynamoDB, PostgreSQL, Elasticsearch, Redis

Infrastructure
AWS ECS, Lambda, S3, CloudFront, API Gateway

DevOps
Docker, Terraform, CodePipeline, CloudWatch

Integrations
Stripe, Google Maps, Twilio, Firebase

Comprehensive Security Model

Security is paramount. "Where's Fluffy?" integrates industry-leading practices to protect user data and ensure platform integrity.

OAuth 2.0 & JWT

Secure user authentication and authorization with JSON Web Tokens.

TLS 1.3 & AES-256

Data encrypted in transit (TLS 1.3) and at rest (AES-256) for maximum protection.

Role-Based Access

Granular control over user permissions, limiting access to sensitive functions.

GDPR & PCI DSS

Compliance with global data privacy and payment card industry standards.

Biometric Login

Enhanced mobile security with fingerprint and facial recognition options.

🔐 Security Highlights

Data Encryption: AES-256 at rest, TLS 1.3 in transit
Authentication: OAuth 2.0 with JWT tokens and MFA
Compliance: PCI DSS for payments, GDPR for data privacy
Monitoring: AWS GuardDuty, Security Hub, and custom anomaly detection

Key Quality Attributes

Our design prioritizes critical quality attributes, ensuring a reliable, responsive, and secure experience for all users.

99.9%

Availability

Targeting high uptime with resilient architecture and redundant systems.

200ms

Performance

API response times under 200 milliseconds for a smooth user experience.

Zero

Security Breaches

Robust security measures aimed at preventing any unauthorized access or data loss.

Multi-city

Scalability

Designed to expand seamlessly across multiple cities and handle increasing user loads.

Uptime

Trade-off

Prioritizing eventual consistency to ensure maximum system uptime.

Anticipating Changes & Failures

Expected Changes

- **Multi-City Rollout:** Gradual expansion to new geographical regions, requiring adaptable infrastructure.
- **Feature Updates (AI Matching):** Integration of machine learning for advanced pet identification and matching based on images and descriptions.

Failure Modes & Mitigation

- **Downtime:** Multi-Availability Zone (AZ) Deployment: Services are distributed across multiple physically separate locations within an AWS region.
- **Data Loss:** Automated Backups & Replication: Regular, incremental backups to S3 and cross-region replication for critical data.
- **Fraudulent Reports:** AI-Powered Photo Verification: Automated analysis of uploaded images to detect duplicates or manipulated content, complemented by human moderation.

Platform Limitations & Constraints

Technical Challenges
- GPS accuracy issues in indoor/urban environments (10-50m variance)
- AI photo verification 15-20% false positive rate
- Mobile dependency excludes non-smartphone users

Business Constraints
- Low repeat usage per customer (infrequent pet loss)
- Market size ceiling in smaller metropolitan areas
- Economic sensitivity affects reward amounts

Operational Reality
- Community engagement follows 90-9-1 participation rule
- Manual moderation doesn't scale with growth
- Cross-regional coverage gaps during expansion

Mitigation: Conservative projections, hybrid automation, strategic partnerships, flexible reward structures

Cost Metrics & Unit Economics

Monthly Infrastructure Costs

Phase 1 (Single City)	$4,500
Phase 2 (Enhanced Features)	$10,000
Phase 3 (Multi-City)	$28,000
Phase 4 (National Scale)	$75,000

Cost per Active User

Phase 1 (1K users)	$4.50/month
Phase 2 (5K users)	$2.00/month
Phase 3 (25K users)	$1.12/month
Phase 4 (100K users)	$0.75/month

Key Financial Metrics

Revenue per Recovery

$15

3% fee on $500 avg reward

Net Profit Margin

67%

$10 profit per recovery

Break-even (Phase 1)

450

successful recoveries/month

Link: Cost Metric Formula

Implementation Roadmap

1

Phase 1: MVP Launch

Single city deployment with core features

2

Phase 2: AI Integration

Advanced matching algorithms and fraud detection

3

Phase 3: Multi-City

Regional expansion with localized features

4

Phase 4: National Scale

Nationwide platform with enterprise partnerships

Performance Targets

API Response: <200ms

Uptime: 99.9%

Concurrent Users: 10,000 per city

Image Upload: <5 seconds

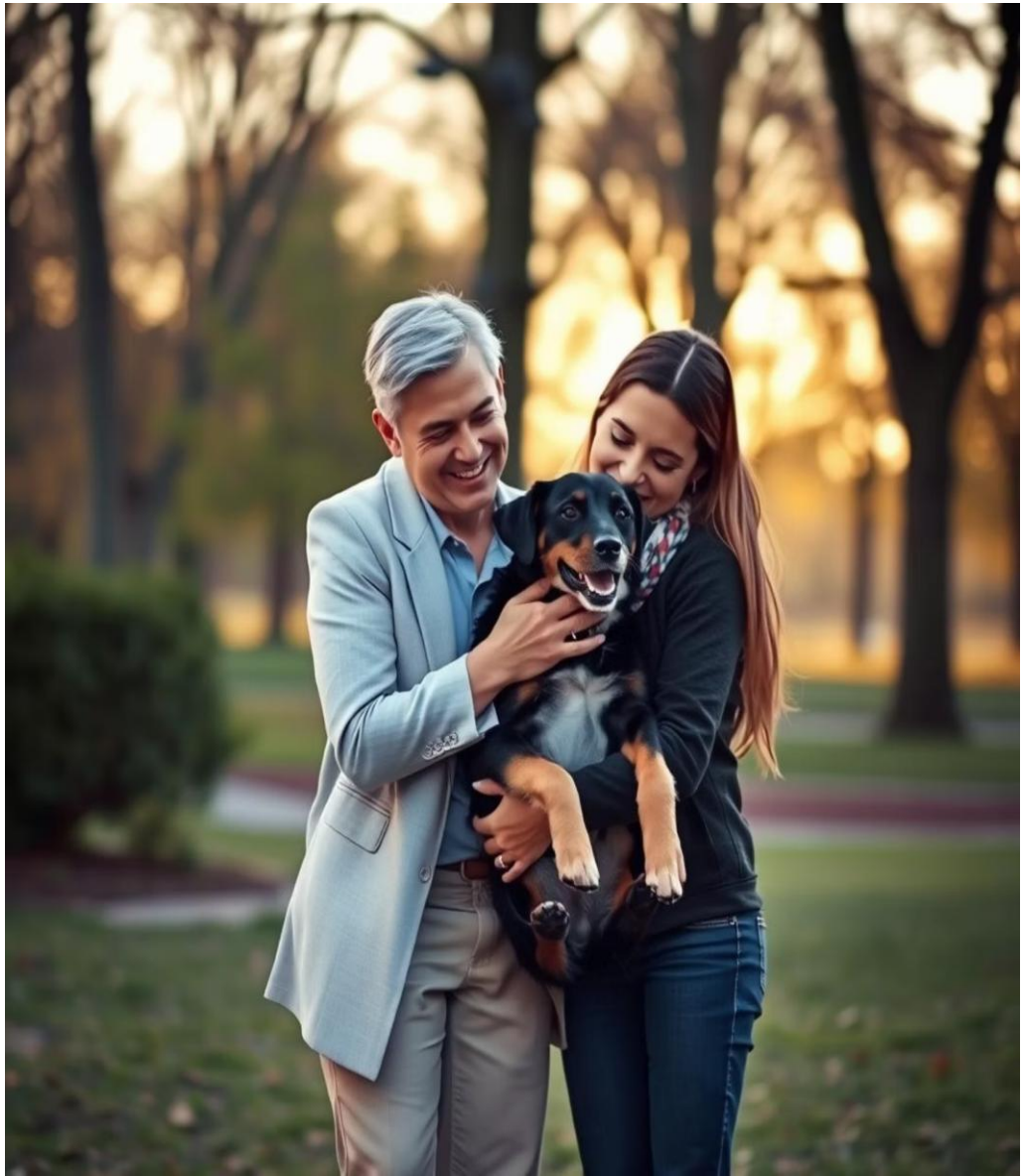
Push Notifications: <30 seconds

Recovery Rate: 70% within 7 days

Summary: Where's Fluffy

"Where's Fluffy" stands as a scalable, secure, and real-time solution for modern pet recovery. By leveraging cutting-edge technology and a community-driven approach, we are committed to:

> Empowering communities through technology.
>
> Saving pets. Reuniting families.



Thank You